

Written Report 3

6.419x Data Analysis Module 3

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22 de julio de 2026

Problem 1: Suggesting Similar Papers

Q1. (2 points) How does the time complexity of your solution involving matrix multiplication in part (a) compare to your friend's algorithm?

Solution.

The matrix multiplication approach computes the cocitation matrix as $A^T A$. Using standard matrix multiplication, its time complexity is $O(n^3)$, which is the same worst-case complexity as my friend's algorithm. The main difference is practical: matrix multiplication is more concise and can take advantage of highly optimized linear algebra libraries. It can also be faster when the citation matrix is sparse, because sparse matrix multiplication avoids many operations involving zeros. Therefore, although both methods have the same worst-case complexity, the matrix-based solution is generally simpler and more efficient in practice.

Q1. (3 points) Bibliographic coupling and cocitation can both be taken as an indicator that papers deal with related material. However, they can in practice give noticeably different results. Why? Which measure is more appropriate as an indicator for similarity between papers?

Solution.

Bibliographic coupling and cocitation capture similarity from different directions. Bibliographic coupling compares the references cited by two papers, so it reflects the knowledge base they share at the time they are published. Cocitation instead depends on later papers citing both documents together, so it reflects how the scientific community subsequently perceives their relationship. As a result, two papers may cite similar literature but never be cocited, or they may rely on different references and still become strongly cocited because later research connects their contributions.

Neither measure is always superior. Bibliographic coupling is more appropriate for recent papers because it can be computed immediately and does not require waiting for future citations. Cocitation is often more informative for older papers because it incorporates the judgment of

later researchers and can reveal relationships that were not obvious from the original reference lists. Therefore, the better measure depends on the objective: bibliographic coupling is useful for contemporaneous similarity, while cocitation is better for measuring similarity as recognized over time.

Problem 2: Investigating a time-varying criminal network

(2 points) Q1. Observe the plot you made in Part (a) Question 1. The number of nodes increases sharply over the first few phases then levels out. Comment on what you think may be causing this effect. Based on your answer, should you adjust your conclusions in Part (b) Question 5?

Solution.

The sharp increase in nodes during the first phases probably does not represent a real expansion of the criminal organization. More likely, the investigators gradually identified new participants and expanded the wiretap network as the investigation progressed. Once most relevant actors had been discovered, the number of observed nodes naturally leveled out. Therefore, the conclusions from Part (b), Question 5 should be interpreted with some caution. Assigning zero centrality to actors before they were detected penalizes late-appearing players and favors actors observed from the beginning, such as n1. Thus, the rankings measure both centrality and duration of observation. A useful comparison would also calculate each player's mean only over the phases in which that player was present.

(4 points) Q2. In the context of criminal networks, what would each of these metrics (including degree, betweenness, and eigenvector centrality) teach you about the importance of an actor's role in the traffic? In your own words, could you explain the limitations of degree centrality? In your opinion, which one would be most relevant to identify who is running the illegal activities of the group?

Solution.

Degree centrality measures how many direct contacts an actor has:

$$C_D(i) = \frac{k_i}{n-1}.$$

In this criminal network, a high degree could indicate that a person is very active in communication and directly coordinates with many participants. Such an actor may be a manager, organizer, or simply someone responsible for routine operational tasks.

Betweenness centrality measures how often an actor lies on the shortest communication paths between other actors. A player with high betweenness may act as a broker between otherwise separated parts of the organization, for example connecting suppliers, transporters, and investors. Removing this person could seriously disrupt communication. However, a broker is not necessarily the leader; the role may belong to an intermediary whose job is to connect groups.

Eigenvector centrality considers both the number and the importance of an actor's contacts. A player receives a high score when connected to other highly central players. It can therefore reveal people located near the influential core of the organization, even when they do not communicate directly with everyone.

A major limitation of degree centrality is that it only counts direct connections and treats all neighbors as equally important. It does not distinguish between contact with a leader and contact with a peripheral participant. It also ignores whether an actor connects different groups. Moreover, leaders of criminal organizations may intentionally maintain few direct contacts to avoid detection, so a low degree does not necessarily imply little authority.

In my opinion, eigenvector centrality is the most relevant metric for identifying who may be running the illegal activities, because leaders are likely to be connected to other important organizers and coordinators. Nevertheless, this result should be interpreted together with betweenness centrality and contextual police information. Centrality measures communication structure, not formal authority, and a careful leader may delegate communications to lieutenants.

(3 points) Q3. In real life, the police need to effectively use all the information they have gathered, to identify who is responsible for running the illegal activities of the group...

Solution.

I define an important player as someone who remains connected to other influential actors and can coordinate or control communication between different parts of the network. Therefore, I would consider both eigenvector and betweenness centrality. The quantitative results identify n1 as the most important actor: he has the highest mean in both metrics, which agrees with his known role as the mastermind. Players n3 and n12 are also central. n3 is highly connected to the influential core and acts as Serero's principal lieutenant, while n12 has high betweenness and likely connects the cocaine operation with the rest of the organization. Player n85 also has high eigenvector centrality, suggesting an important financial role. In contrast, n33, n80, and n16 appear structurally peripheral, although their low scores may reflect specialized roles or participation in only a few phases.

(3 points) Q4. The change in the network from Phase X to X+1 coincides with a major event that took place during the actual investigation. Identify the event and explain how the change in centrality rankings and visual patterns, observed in the network plots above, relates to said event.

Solution.

The event was the first police seizure, which occurred in Phase 4: approximately 300 kg of marijuana, valued at \$2.5 million, was confiscated. After this loss, the organization changed its trafficking strategy from importing marijuana from Morocco through Spain to importing cocaine from Colombia through the United States.

The transition from Phase 4 to Phase 5 shows this reorganization. Player n1 remains the dominant hub and has the highest centrality, suggesting that Serero continued to direct the network despite the seizure. However, n12 becomes much more prominent in Phase 5, ranking second in degree, betweenness, and eigenvector centrality. This is significant because Ernesto Morales (n12) was the principal organizer of the cocaine imports and the intermediary between the Colombian suppliers and Serero's organization.

Visually, Phase 4 is centered strongly around n1, with n89 forming an important secondary branch. In Phase 5, a new cluster develops around n12, while the relative importance of actors such as n83 and n89 declines. Therefore, the seizure did not destroy the network; instead, it caused the organization to adapt by changing its product, route, and operational personnel.

(4 points) Q5. Describe the coarse pattern(s) you observe as the network evolves through the phases. Does the network evolution reflect the background story?

Solution.

At a coarse level, the network first expands rapidly, from 15 actors in Phase 1 to 33 in Phase 3, and then fluctuates rather than growing continuously. Actors repeatedly enter and leave the observed network, while the number of edges also rises and falls. This suggests a flexible organization that reorganizes its communication instead of maintaining a fixed membership.

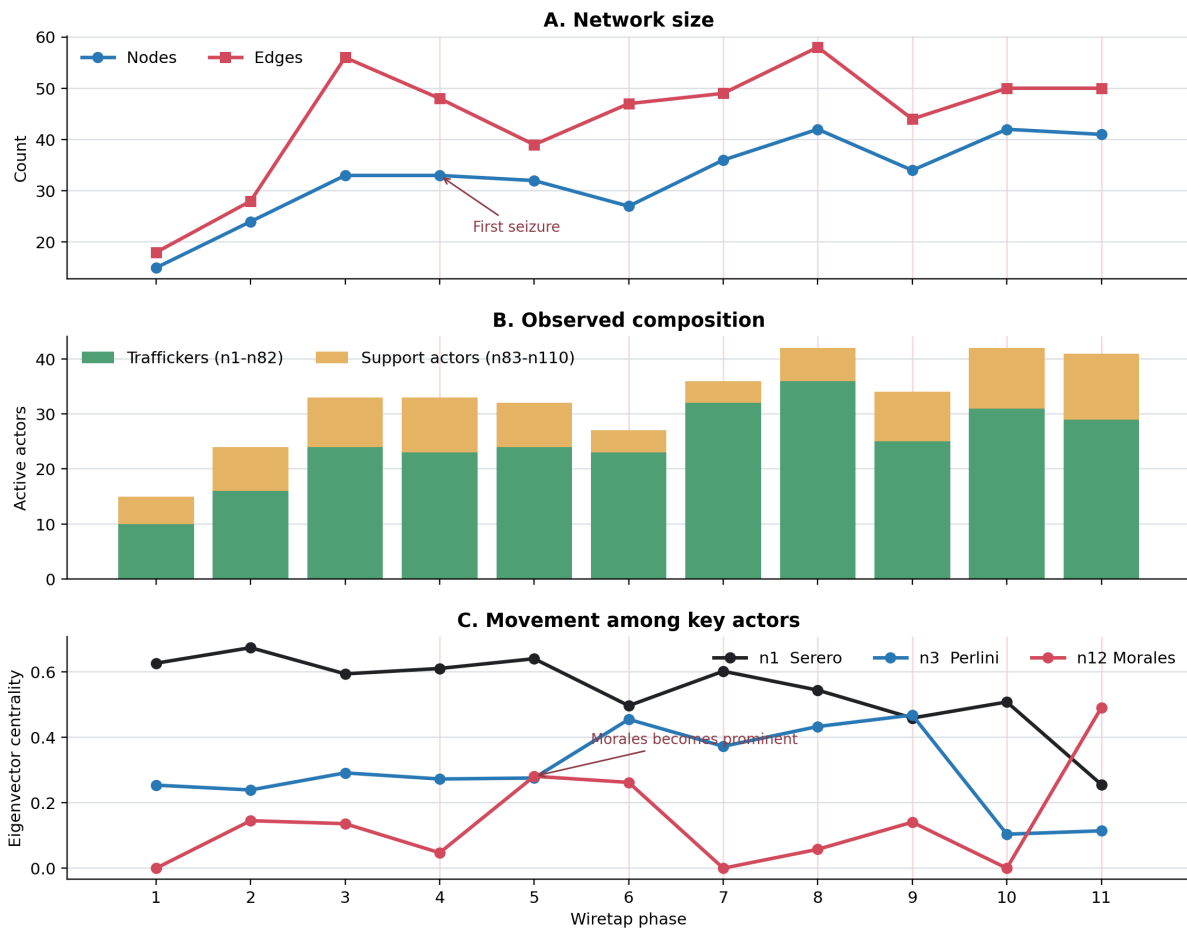


Figure 1: Evolution of network size, composition, and key actors across the eleven CAVIAR phases. Light red lines indicate phases with police seizures.

The evolution generally reflects the background story. Serero (n1) remains the main structural hub through most phases, indicating continuity in leadership despite repeated seizures. After the first seizure in Phase 4, Morales (n12), the organizer of cocaine imports, becomes considerably more central in Phases 5 and 6. This is consistent with the reported transition from the Morocco–Spain marijuana route toward cocaine imported through the United States. Perlini (n3) also gains importance in later phases, at times approaching or exceeding n1, which may indicate greater delegation to Serero’s lieutenant.

The later phases contain more traffickers and, especially in Phases 10 and 11, more supporting actors. The network does not show a permanent collapse after seizures because arrests occurred only at the end. Instead, it repeatedly changes composition and redistributes centrality. The

renewed prominence of n12 in Phase 11 further suggests adaptation toward the cocaine operation.

(2 points) Q6. Are there other actors that play an important role but are not on the list of investigation (i.e., actors who are not among the 23 listed above)? List them, and explain why they are important.

Solution.

Yes. Outside the police's 23-player list, n2 and n9 appear important because they have the highest mean eigenvector and degree centralities among unlisted actors and remain active across many phases. Player n41 has the highest mean betweenness in this group, while n14 ranks second. They may represent influential associates or communication brokers overlooked by the qualitative list.

(2 points) Q7. What are the advantages of looking at the directed version vs. undirected version of the criminal network?

A directed network preserves who initiated each communication and who received it, information that is lost when every relationship is treated as undirected. In-degree centrality identifies actors contacted by many others. These actors may receive reports, requests, money-related information, or approval requests. Out-degree centrality identifies actors who contact many others and may therefore distribute instructions, coordinate operations, or actively gather information. Comparing the two can distinguish leaders, coordinators, information collectors, and intermediaries even when they have similar total degrees.

Assuming $A_{ij} = 1$ represents a message from i to j , left-eigenvector centrality rewards actors receiving links from already influential senders. It can identify important information destinations or authorities. Right-eigenvector centrality rewards actors sending links toward influential recipients, revealing actors connected outward to the network's important core.

Thus, direction helps distinguish influence received from influence exercised. The undirected graph is simpler and captures general association, but it may incorrectly make a frequent sender and an important recipient appear structurally equivalent.

(4 points) Q8. Using this, what relevant observations can you make on how the relationship between n1 and n3 evolves over the phases. Can you make comparisons to your results in Part (g)?

HITS defines a good hub as an actor who sends communications to strong authorities, while a good authority receives communications from strong hubs. I computed both normalized scores for every actor in every directed phase. During Phases 1–5, n1 clearly dominates as a hub, with scores between 0.360 and 0.673, compared with approximately 0.084–0.094 for n3. However, n3 has a higher authority score than n1 throughout these phases. This is consistent with Serero actively coordinating the network and Perlini receiving communications as the lieutenant who executes his instructions.

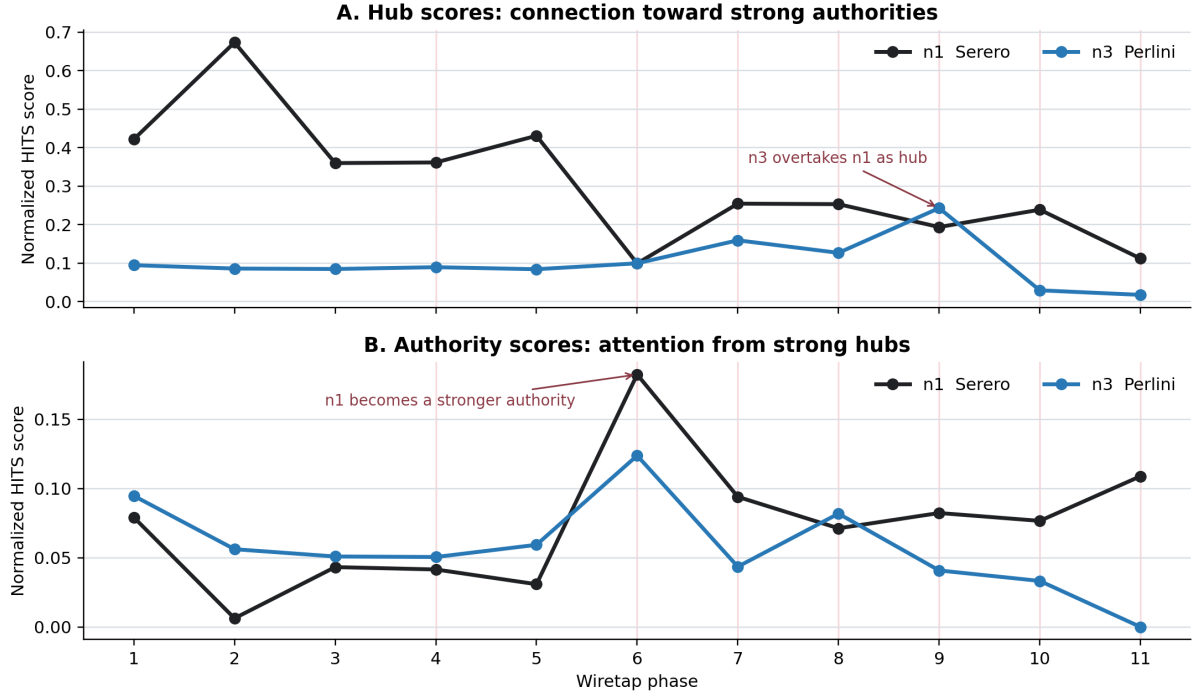


Figure 2: Evolution of the normalized HITS scores of n1 and n3. Light red lines identify phases containing police seizures.

Their relationship changes around Phase 6, when their hub scores become almost identical at 0.099, while n1’s authority rises sharply to 0.182. The clearest reversal occurs in Phase 9: n3’s hub score reaches 0.243, exceeding n1’s 0.194, whereas n1 remains the stronger authority (0.082 versus 0.041). This suggests that Perlini temporarily assumed more responsibility for contacting influential actors, while Serero increasingly became a destination for communications. In Phases 10–11, n3’s scores decline sharply, but n1 retains a meaningful hub and authority role.

These findings support Part (g): leadership remains centered on n1, but operational coordination is periodically delegated to n3 as the organization adapts. Across all phases, n1 also has the highest mean hub and authority scores, followed by n3. Wallace Lee (n85) ranks third on both averages, reinforcing the importance of his financial role identified in Part (e). Nevertheless, HITS measures communication structure rather than formal command, and comparisons across phases are affected by changes in network composition.

Project

Research Question and Motivation

Social networks are often organized around similarity. People who share educational, professional, political, religious, or geographic backgrounds may be more likely to interact and to form tightly connected groups. These relationships can generate *bonding social capital*, characterized by cohesion, trust, and repeated interaction within relatively homogeneous groups. At the same time, individuals whose contacts come from different social contexts may play a different role by connecting otherwise separated communities, creating what is commonly described as *bridging social capital*.

This project asks:

Is social similarity associated with greater cohesion within groups, while social diversity is associated with greater brokerage between groups in Facebook ego-networks?

The question is relevant because cohesion and brokerage represent two distinct forms of social position. A person may be well integrated within a dense and homogeneous group, but have limited access to information outside that group. In contrast, a person connected to socially diverse contacts may occupy a less cohesive local environment while gaining access to different communities, information, and opportunities.

We expect nodes with more socially similar neighborhoods to display higher local clustering, since their contacts are more likely to know one another. Conversely, we expect nodes with more socially diverse neighborhoods to have higher betweenness centrality and a higher participation coefficient, as they are more likely to connect different communities. We also expect these relationships to remain after accounting for degree, since having many connections does not necessarily imply connecting socially distinct groups.

Data and Network Construction

The analysis uses the Facebook ego-network dataset provided by the Stanford Network Analysis Project (SNAP). The data were collected from survey participants through a Facebook application and contain friendship ties, anonymized profile features, manually reported social circles, and separate ego-networks centered around each survey participant.

A node represents an anonymized Facebook user and an edge represents a Facebook friendship between two users. Since Facebook friendships are mutual, each ego-network is modeled as an undirected graph. The network is also treated as unweighted because the data indicate whether a friendship exists, but do not contain information about the frequency, intensity, or duration of the relationship.

The raw archive contains ten ego-networks and 4,177 node–ego records, corresponding to 4,039 distinct anonymized user IDs. The distinction is important because the same user may appear in more than one ego-network and is therefore treated as a separate observation within each network. The raw `.edges` files contain 170,174 line records because each undirected alter–alter relationship appears in both orientations. After removing these duplicates, the files contain 85,087 unique alter–alter ties. The ego node and its documented connections to all alters were then added to each network, producing 89,254 friendship ties across the ten constructed ego-networks.

All ten resulting networks are connected, contain no isolated nodes, and therefore coincide with their largest connected components. Network size varies considerably: the smallest ego-

network contains 60 nodes and 205 edges, while the largest contains 1,046 nodes. The number of edges ranges from 205 to 30,780. Table 1 summarizes the structure of each network.

Table 1: Construction and basic structure of the Facebook ego-networks

Ego ID	Nodes	Edges	Density	Mean degree	Components
0	348	2,866	0.047	16.47	1
107	1,046	27,794	0.051	53.14	1
348	228	3,419	0.132	29.99	1
414	160	1,852	0.146	23.15	1
686	171	1,826	0.126	21.36	1
698	67	336	0.152	10.03	1
1684	793	14,816	0.047	37.37	1
1912	756	30,780	0.108	81.43	1
3437	548	5,360	0.036	19.56	1
3980	60	205	0.116	6.83	1

Notes: Edges are undirected and counted within ego-network. The ego node and its ties to the alters are included. Density is calculated as $2m/[n(n-1)]$. The same anonymized user ID may appear in more than one ego-network.

Profile information is represented through binary feature vectors. The features describe broad social dimensions such as birthday, education, gender, hometown, languages, last name, locale, location, and work. The underlying values have been anonymized. Thus, the data may show that two users share an educational institution or location, but not reveal the identity of that institution or location.

Moreover, the interpretation of anonymized feature columns is specific to each ego-network. A feature value in one network cannot be assumed to represent the same social category in another network. For this reason, pairwise similarity and neighborhood diversity are always calculated within ego-network. The broad feature categories are comparable across networks, but their anonymized values are not pooled directly. Subsequent pooled models retain the ego-network identifier and include ego-network fixed effects.

The analytical samples are defined according to the information required for each part of the analysis. Nodes without usable profile attributes remain in the graph when structural statistics are calculated, since removing them would alter paths, degrees, and community structure. They are excluded only from analyses requiring social similarity or diversity. Nodes with fewer than two neighbors are additionally excluded from the cohesion analysis because their local clustering coefficient is not informative. Table 2 reports the resulting samples.

Table 2: Definition of the analytical samples

Sample	Ego-networks	Node-ego records	Excluded	Retained (%)
Raw universe	10	4,177	0	100.00
A: Structural analysis	10	4,177	0	100.00
B: Social similarity	10	4,169	8	99.81
C: Local cohesion	10	4,099	78	98.13
D: Brokerage	10	4,169	8	99.81
E: Main regressions	10	4,099	78	98.13

Notes: Sample A contains all structurally valid nodes. Sample B requires usable attributes and at least one neighbor with usable attributes. Sample C additionally requires degree $k_i \geq 2$. Sample D requires membership in the connected component used for path-based measures. Since all ten ego-networks are connected, this restriction causes no additional loss relative to Sample B. Sample E is the intersection of the variables required for the main regression models. Exclusions are measured relative to the raw universe.

Feature coverage is high: on average, 99.5 % of node–ego records contain usable profile information. Consequently, only eight observations are lost from the social-similarity and brokerage samples. The main cohesion and regression samples contain 4,099 records, retaining 98.13 % of the original node–ego observations.

Several limitations remain. First, the dataset consists of ten sampled ego-networks and is not representative of the full Facebook population. Second, the anonymization of the features prevents identifying the substantive meaning of specific affiliations. Third, profile information may be incomplete or selectively reported. Fourth, a Facebook friendship indicates the existence of a declared relationship but not its strength or level of interaction. Finally, the network is cross-sectional. The analysis can therefore identify associations between social similarity, cohesion, and brokerage, but cannot establish that social similarity causes cohesion or that social diversity causes brokerage.

Methodology

The analysis was conducted at the node level. For each user, we constructed measures of social similarity, local cohesion, and brokerage. The objective was to evaluate whether users embedded in socially homogeneous neighborhoods occupy more cohesive positions, while users connected to socially diverse contacts are more likely to bridge different parts of the network.

Social similarity and neighborhood diversity

Let x_i denote the binary profile-feature vector of user i . Social similarity between two connected users i and j was measured using the Jaccard coefficient:

$$S_{ij} = \frac{|x_i \cap x_j|}{|x_i \cup x_j|}.$$

This measure is appropriate because the Facebook profile features are represented as binary indicators, and the joint absence of a feature should not be treated as evidence of similarity. For each node, we calculated its average similarity with its neighbors:

$$\bar{S}_i = \frac{1}{k_i} \sum_{j \in N(i)} S_{ij},$$

where $N(i)$ is the set of neighbors of node i , and k_i is its degree.

Neighborhood diversity was defined as the complement of average similarity:

$$D_i = 1 - \bar{S}_i.$$

Higher values of D_i therefore indicate that a user is connected to contacts with less similar observed profiles. Because this main diversity measure is the exact complement of average similarity, it should not be interpreted as an independent dimension of social composition. Its purpose is to express the same profile information in a form that is easier to interpret in the brokerage models.

To assess whether the findings depended on this particular construction, the robustness analysis also used category entropy, circle entropy, and the share of ties outside the node’s dominant reported circle as alternative measures of neighborhood diversity. Cosine similarity and similarity measures based on individual feature categories were also examined as alternatives to the main Jaccard measure.

Local cohesion

Local cohesion was measured using the local clustering coefficient:

$$C_i = \frac{2T_i}{k_i(k_i - 1)},$$

where T_i is the number of triangles containing node i . The coefficient measures the proportion of possible connections among a user’s neighbors that are actually observed. A high value indicates that the user’s contacts also tend to be connected to one another, which is consistent with a dense and cohesive local social environment.

The clustering coefficient is preferable to degree as a measure of cohesion because a user may have many contacts who are otherwise unrelated. Degree captures the size of the neighborhood, whereas clustering captures its internal connectedness. Nodes with degree below two were excluded from the cohesion models because the clustering coefficient is not informative for them.

Brokerage and inter-community connectivity

Brokerage was first measured using normalized betweenness centrality:

$$B_i = \frac{1}{(n-1)(n-2)/2} \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}},$$

where σ_{st} is the number of shortest paths between nodes s and t , and $\sigma_{st}(i)$ is the number of those paths that pass through i . Betweenness captures whether a user occupies a structurally strategic position between otherwise separated parts of the network.

Because betweenness is highly skewed and strongly related to degree, the regression analysis used

$$\log(1 + B_i)$$

as the dependent variable and included $\log(1 + k_i)$ as a control.

A second measure of brokerage was the participation coefficient:

$$P_i = 1 - \sum_{c=1}^M \left(\frac{k_{ic}}{k_i} \right)^2,$$

where k_{ic} is the number of links from node i to community c , and M is the total number of detected communities. Values close to zero indicate that most of a node’s ties are concentrated within a single community, while larger values indicate that its ties are distributed more evenly across several communities.

We also calculated the number of detected communities reached by each node. Unlike betweenness, these two measures capture inter-community connectivity directly rather than relying exclusively on shortest paths.

Community detection

Communities were identified separately within each ego-network using the Louvain modularity-maximization algorithm. The same procedure and resolution setting were applied across networks to maintain comparability.

For a partition of the network into communities, modularity is defined as

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \mathbf{1}(g_i = g_j),$$

where A_{ij} is the adjacency matrix, m is the number of edges, and g_i denotes the community assigned to node i . Modularity compares the observed concentration of ties within communities with the concentration expected in a network with the same degree distribution.

The detected communities were interpreted as structural groups rather than as directly observed social categories. They were used to calculate the participation coefficient and the number of communities reached, while the reported Facebook circles were used only in the construction of alternative diversity measures.

Statistical analysis

The first hypothesis was evaluated by estimating the relationship between average social similarity and local clustering:

$$C_i = \alpha + \beta_1 \bar{S}_i + \beta_2 \log(1 + k_i) + \mu_e + \varepsilon_i,$$

where μ_e represents ego-network fixed effects. The bonding hypothesis implies

$$\beta_1 > 0.$$

The second hypothesis was evaluated using:

$$\log(1 + B_i) = \alpha + \gamma_1 D_i + \gamma_2 \log(1 + k_i) + \mu_e + \varepsilon_i,$$

and

$$P_i = \alpha + \delta_1 D_i + \delta_2 \log(1 + k_i) + \mu_e + \varepsilon_i.$$

The bridging hypothesis implies

$$\gamma_1 > 0 \quad \text{and} \quad \delta_1 > 0.$$

The number of communities reached was analyzed using a Poisson count model with the same main predictor, degree control, and ego-network fixed effects.

The fixed effects account for persistent differences across ego-networks, including network size, density, composition, and sampling. Because the sample contains only ten ego-networks, conventional cluster-robust inference may be unreliable. The pooled regressions therefore report HC3 heteroskedasticity-robust standard errors and are complemented by within-ego-network permutation tests.

Spearman rank correlations were also calculated as non-parametric descriptive measures of association. These correlations reduce sensitivity to skewed centrality distributions and extreme observations, but they do not account for degree or differences across ego-networks. They were therefore interpreted jointly with the adjusted regression models rather than as the main inferential results.

Null models and permutation tests

To assess whether the estimated relationships depended on the observed placement of social attributes, we constructed a permutation-based null model. Within each ego-network, complete profile vectors were randomly reassigned across nodes while the friendship structure remained fixed. Social similarity and diversity were then recalculated, followed by the corresponding correlation and regression coefficients.

The empirical two-sided p -value was calculated as

$$p = \frac{1 + \sum_{r=1}^R \mathbf{1}(|T_r| \geq |T_{\text{obs}}|)}{R + 1},$$

where T_{obs} is the observed statistic, T_r is the statistic obtained in permutation r , and $R = 1,000$ is the number of permutations.

This procedure preserves the number of nodes, edges, degree distribution, community structure, and all other structural characteristics of each ego-network. At the same time, it removes the observed association between profile characteristics and network position. It therefore provides a more appropriate benchmark for the hypotheses than a fully randomized Erdős–Rényi network.

Sample selection and robustness

All ten ego-networks were retained because they contained valid friendship data, high profile-feature coverage, and no disconnected components. Nodes without usable profile attributes remained in the graphs for the calculation of structural measures, but were excluded from analyses requiring similarity or diversity. Nodes with degree one were retained for brokerage measures and excluded from clustering regressions.

The pooled models included ego-network fixed effects, while separate estimates were calculated by ego-network to assess heterogeneity. Robustness checks excluded the most central nodes, restricted the sample to larger or more modular networks, and used alternative measures of similarity and diversity.

Some anonymized users appear in more than one ego-network. These observations were treated separately to preserve their local network context, although this may introduce residual dependence across observations.

Results

Descriptive Network Structure

The ten ego-networks differ substantially in size and connectivity. As shown in Figure 3, the number of nodes ranges from 60 to 1,046, while the number of edges ranges from 205 to 30,780. Network size and density do not move together: some of the smallest ego-networks are among the densest, whereas several of the largest networks have densities below 0.06. This is expected because the number of possible ties grows quadratically with the number of nodes.

Despite these differences, all networks exhibit relatively high local cohesion. Mean clustering ranges from 0.537 to 0.750, indicating that a large proportion of each user’s contacts are also connected to one another. The networks nevertheless vary in the strength of their community structure. Louvain modularity ranges from 0.298 to 0.623, and the algorithm identifies between four and nine communities per ego-network. Thus, the data contain both densely connected local neighborhoods and meaningful variation in the extent to which those neighborhoods are separated into larger social groups.

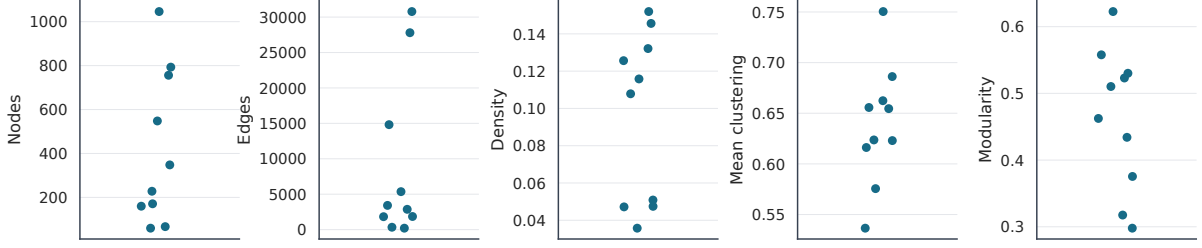


Figure 3: Structural characteristics of the Facebook ego-networks

Notes: Each point represents one of the ten ego-networks. Edges are undirected and include the connections between each ego and its alters. Clustering is the mean local clustering coefficient. Modularity corresponds to the selected Louvain partition.

Table 3 summarizes the node-level variables used in the analysis. The average node has approximately 43 connections, although the degree distribution is strongly right-skewed: the median is 25 and the maximum is 1,045. Average Jaccard similarity with neighboring users is 0.234, implying an average diversity complement of 0.766. These measures should not be interpreted as absolute levels of social similarity across Facebook, since they are constructed from the anonymized features available within each ego-network.

The mean local clustering coefficient is 0.609, consistent with the high levels of local cohesion observed at the network level. Brokerage measures are considerably more unequal. Median normalized betweenness is only 0.000088, while its maximum reaches 0.865, indicating that a small number of nodes occupy unusually important positions on shortest paths. The mean participation coefficient is 0.225, and the typical node reaches approximately three detected communities. These distributions reinforce the need to distinguish popularity from brokerage and to control for degree in the main analysis.

Table 3: Node-level descriptive statistics

Variable	<i>N</i>	Mean	SD	Min.	P25	Median	P75	Max.
Degree	4,177	42.736	51.563	1.000	11.000	25.000	55.000	1,045.000
Mean Jaccard similarity	4,169	0.234	0.077	0.000	0.187	0.229	0.276	0.833
Diversity complement	4,169	0.766	0.077	0.167	0.724	0.771	0.813	1.000
Local clustering	4,177	0.609	0.212	0.000	0.471	0.600	0.754	1.000
Betweenness centrality	4,177	0.00219	0.03555	0.00000	0.00002	0.00009	0.00031	0.86517
Participation coefficient	4,177	0.225	0.203	0.000	0.050	0.163	0.375	0.839
Communities reached	4,177	2.609	1.022	1.000	2.000	3.000	3.000	9.000
Constraint	4,177	0.142	0.178	0.004	0.040	0.076	0.167	1.000
Effective size	4,177	18.689	30.857	1.000	3.857	11.082	25.298	993.806

Notes: The unit of observation is a node within an ego-network. Jaccard similarity is the average similarity between a node and its neighbors. The diversity complement is defined as one minus average Jaccard similarity. Betweenness is normalized within each ego-network. The participation coefficient measures how evenly a node’s ties are distributed across detected communities. Constraint measures how concentrated and redundant a node’s connections are, with higher values indicating fewer brokerage opportunities. Effective size approximates the number of non-redundant contacts in the node’s neighborhood, with larger values indicating greater access to structural holes.

Social Similarity and Local Cohesion

We begin by examining whether users embedded in more socially similar neighborhoods also occupy more cohesive local positions. Figure 4 plots average Jaccard similarity against the local clustering coefficient. At the descriptive level, the relationship is weak. Most observations are

concentrated between 0.15 and 0.30 in average similarity, while clustering spans a much wider range. The binscatter suggests a shallow U-shaped pattern rather than a strong monotonic trend, which already indicates that social similarity alone is not enough to summarize local cohesion.

This weak unconditional relationship is confirmed by the pairwise correlation. The Spearman correlation between average similarity and clustering is positive but very small, $\rho = 0.019$, and statistically insignificant ($p = 0.216$). In other words, when nodes are compared without accounting for other structural characteristics, more socially similar neighborhoods do not appear systematically more clustered.

However, this descriptive comparison masks an important confounding factor: degree. Nodes with many neighbors have mechanically different opportunities to form triangles than nodes with only a few contacts. Once we control for degree and include ego-network fixed effects, the relationship becomes positive and statistically significant. In the main regression, the coefficient on average similarity is 0.079 ($p = 0.041$). This implies that, conditional on network-specific factors and neighborhood size, nodes whose contacts are more socially similar tend to exhibit higher local clustering.

Substantively, this result provides qualified support for the bonding hypothesis. The unconditional association between similarity and cohesion is close to zero, but after adjusting for degree and cross-network differences, greater social similarity is associated with more tightly knit local neighborhoods. The magnitude is not large, so the evidence should not be read as showing that similarity fully determines cohesion. Rather, it suggests that social resemblance is one component of local closure, once the structural opportunities to form ties are taken into account.

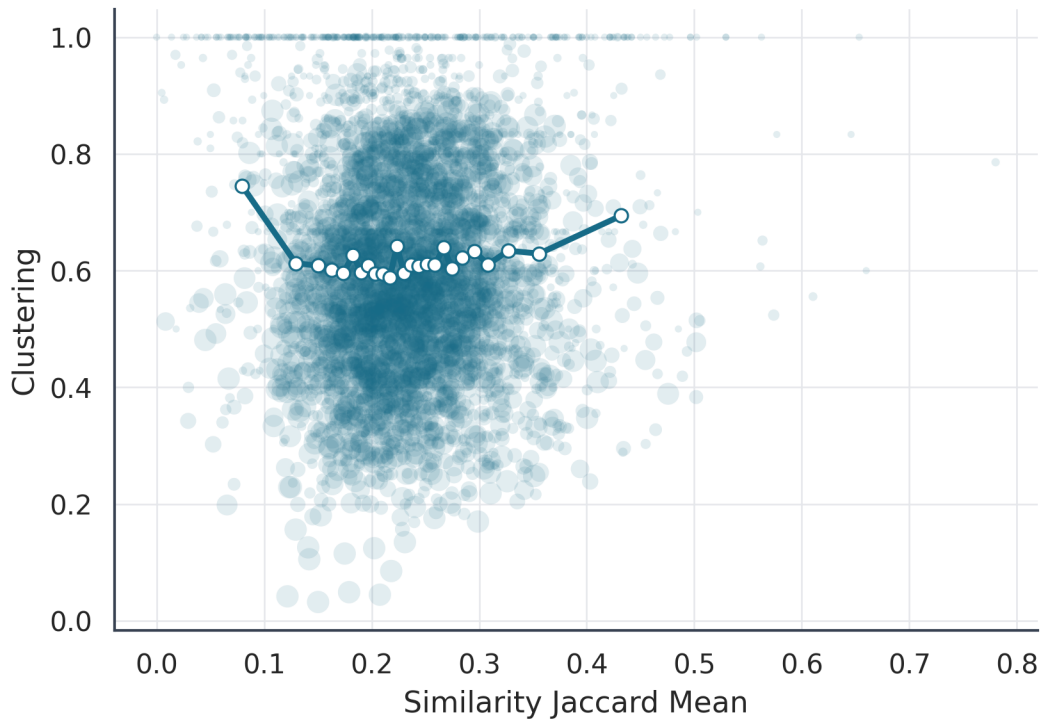


Figure 4: Average social similarity and local clustering

Notes: Each point is a node within an ego-network. The horizontal axis reports the mean Jaccard similarity between the node and its neighbors. The vertical axis reports the local clustering coefficient. The connected markers summarize the binned relationship. Marker size reflects degree.

Table 4: Social similarity and local cohesion

	Estimate	<i>p</i> -value
Spearman correlation: similarity vs. clustering	0.019	0.216
Main regression coefficient on similarity	0.079	0.041

Notes: The first row reports the unconditional Spearman rank correlation between average Jaccard similarity and the local clustering coefficient. The second row reports the coefficient on similarity from the main regression of clustering on similarity, controlling for $\log(1 + \text{degree})$ and including ego-network fixed effects. Full results are reported in the main regression table.

Social Diversity and Brokerage

We next examine whether socially diverse neighborhoods are associated with brokerage across different parts of the network. Diversity is defined as one minus average Jaccard similarity, so higher values indicate that a node is connected to less socially similar contacts.

The relationship with betweenness is positive but weak in the raw data. Figure 5 shows that most nodes have betweenness close to zero, while a small number occupy unusually central positions. The Spearman correlation is therefore small and insignificant ($\rho = 0.011$, $p = 0.475$). However, after controlling for degree and ego-network fixed effects, diversity is positively associated with $\log(1 + \text{betweenness})$, with a coefficient of 0.017 ($p = 0.003$).

The relationship is clearer for inter-community brokerage. As shown in Figure 6, nodes with more diverse neighborhoods tend to distribute their ties across a broader set of communities. In the adjusted model, diversity is positively associated with the participation coefficient ($\beta = 0.292$, $p < 0.001$) and with the number of communities reached ($\beta = 0.624$, $p < 0.001$).

These results support the bridging hypothesis. Social diversity is only weakly related to shortest-path brokerage, but it is more clearly associated with connecting multiple communities. Importantly, these relationships remain after controlling for degree, suggesting that brokerage is not simply a consequence of having more connections.

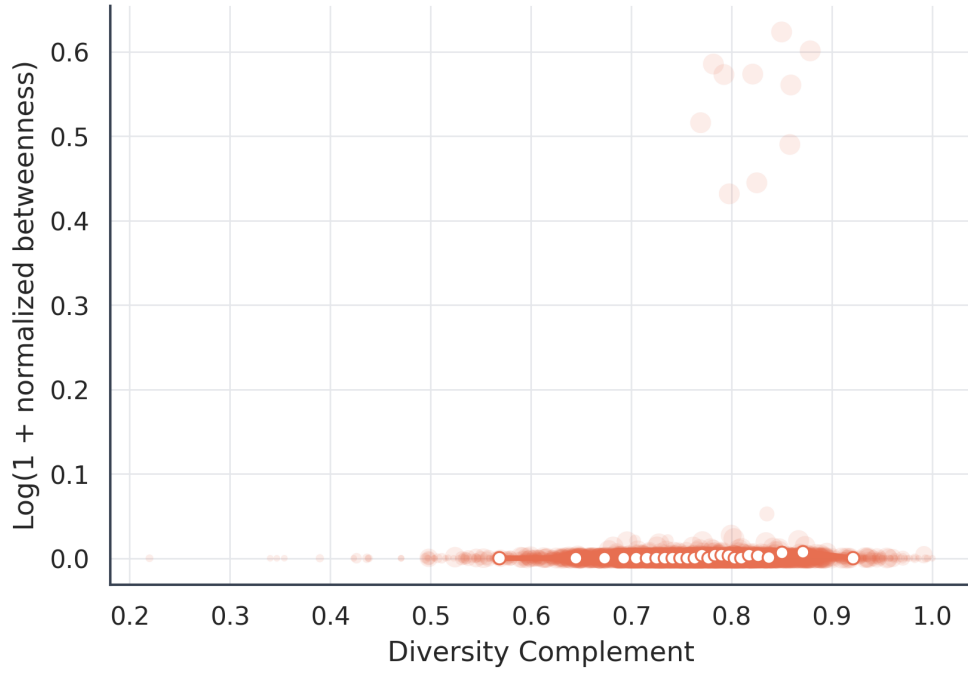


Figure 5: Social diversity and betweenness centrality

Notes: Diversity is defined as one minus average Jaccard similarity. The vertical axis reports $\log(1 + \text{normalized betweenness})$. Connected markers show binned means, and marker size reflects degree.

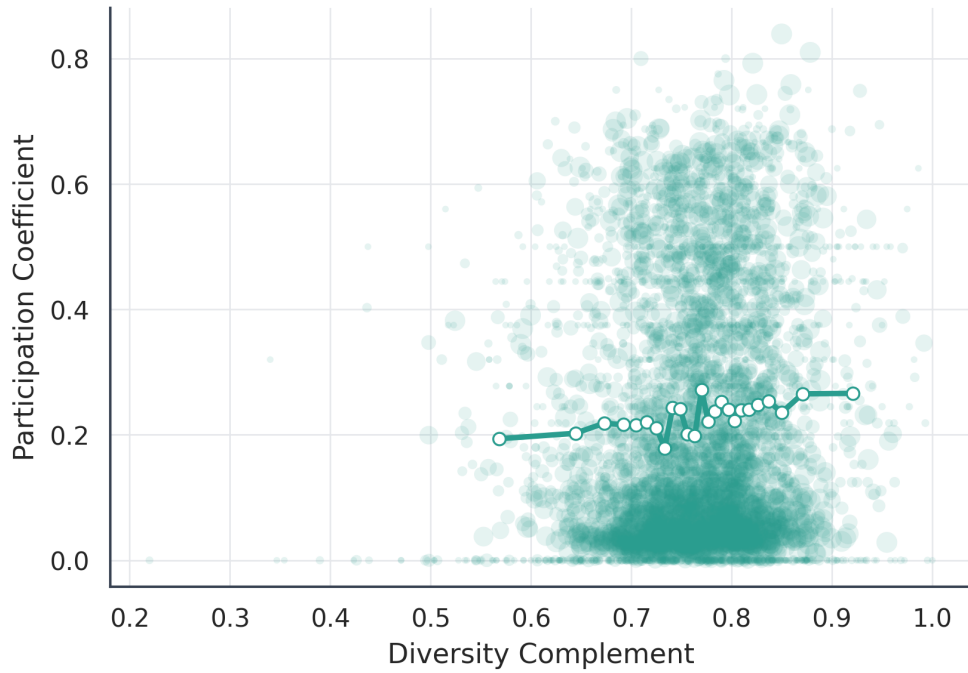


Figure 6: Social diversity and participation across communities

Notes: The participation coefficient increases when a node's ties are distributed more evenly across detected communities. Connected markers show binned means, and marker size reflects degree.

Table 5: Main adjusted results

Outcome	Main predictor	Coefficient	p -value
Local clustering	Social similarity	0.079	0.041
Log betweenness	Social diversity	0.017	0.003
Participation coefficient	Social diversity	0.292	< 0.001
Communities reached	Social diversity	0.624	< 0.001

Notes: All models control for $\log(1 + \text{degree})$, include ego-network fixed effects, and report HC3 heteroskedasticity-robust standard errors. The communities-reached specification is estimated using a Poisson count model.

Null Models and Permutation Tests

We use permutation tests to assess whether the estimated relationships depend on the observed placement of social attributes in the network. Within each ego-network, profile vectors are randomly reassigned across nodes while the friendship structure remains fixed. Similarity, diversity, and the corresponding statistics are then recalculated over 1,000 permutations.

The observed Spearman coefficients lie in the upper tail of their respective null distributions, as shown in Figure 7. However, the two-sided permutation tests for these unconditional correlations are not statistically significant. This occurs because the null distributions are centered below zero, so many permuted coefficients have a larger absolute magnitude than the small positive observed coefficients.

The regression-based tests provide clearer evidence. The adjusted similarity–clustering coefficient is not statistically distinguishable from its permutation distribution ($p_{\text{perm}} = 0.227$). In contrast, the diversity–betweenness coefficient is significant at the 5 % level ($p_{\text{perm}} = 0.017$), while the diversity–participation coefficient is strongly significant ($p_{\text{perm}} = 0.001$). The null-model evidence therefore supports the bridging results, especially the relationship between social diversity and connections across communities, but provides weaker evidence for the bonding hypothesis.

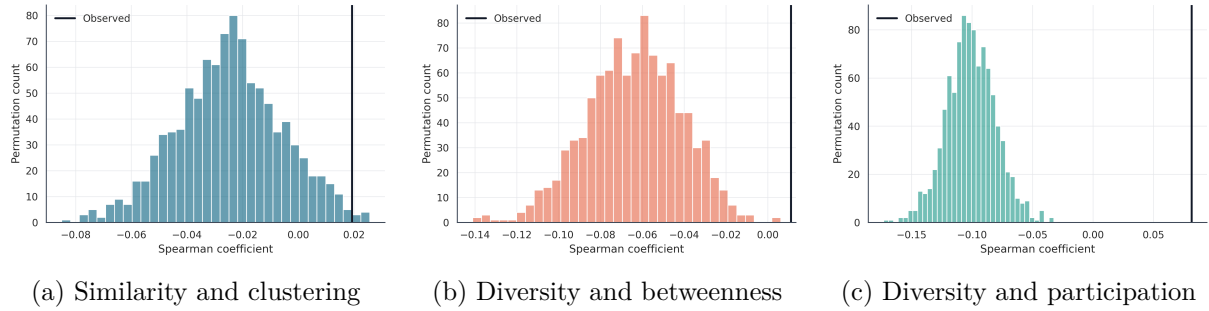


Figure 7: Null distributions of permutation-based Spearman coefficients

Notes: Each histogram reports the distribution of Spearman coefficients obtained after randomly reassigning profile vectors within ego-networks while holding the friendship structure fixed. The vertical line marks the observed coefficient in the original data.

Table 6: Permutation tests for the adjusted relationships

Relationship	Observed coefficient	Empirical p -value
Similarity–clustering	0.079	0.227
Diversity–betweenness	0.017	0.017
Diversity–participation	0.292	0.001

Notes: Empirical two-sided p -values are based on 1,000 within-ego-network permutations. The reported coefficients correspond to the adjusted regression models with degree controls and ego-network fixed effects. The histograms in Figure 7, by contrast, display the permutation distributions of the unconditional Spearman correlations.

Robustness and Heterogeneity

The main results are broadly stable across alternative samples and measures. As shown in Figure 8, the positive association between diversity and participation remains large and statistically significant in every specification. The diversity–betweenness result is also generally positive, although it becomes less precise after excluding the nodes with the highest degree. The similarity–clustering coefficient is more sensitive: it remains positive across specifications, but loses conventional statistical significance when the most central nodes are removed.

The alternative diversity measures lead to a similar conclusion. Circle entropy and the share of ties outside the dominant circle are positively associated with both betweenness and participation. Category entropy is also positively related to participation, although its relationship with betweenness is negative. The bridging result is therefore strongest when brokerage is defined as connecting multiple communities rather than occupying shortest paths.

Figure 9 shows substantial variation across ego-networks. The similarity–clustering relationship is positive in some networks and negative in others. Diversity is more consistently associated with participation, but the magnitude still varies considerably. These differences suggest that the relationship between social composition and network position depends partly on the structure of each local network.

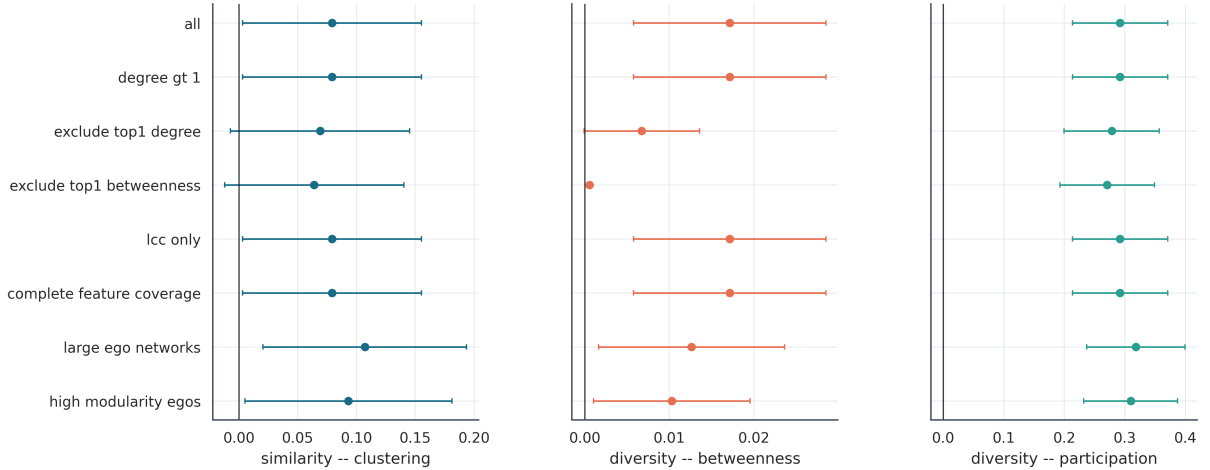


Figure 8: Robustness across alternative analytical samples

Notes: Points report regression coefficients and horizontal lines report 95 % confidence intervals. All models control for $\log(1 + \text{degree})$ and include ego-network fixed effects.

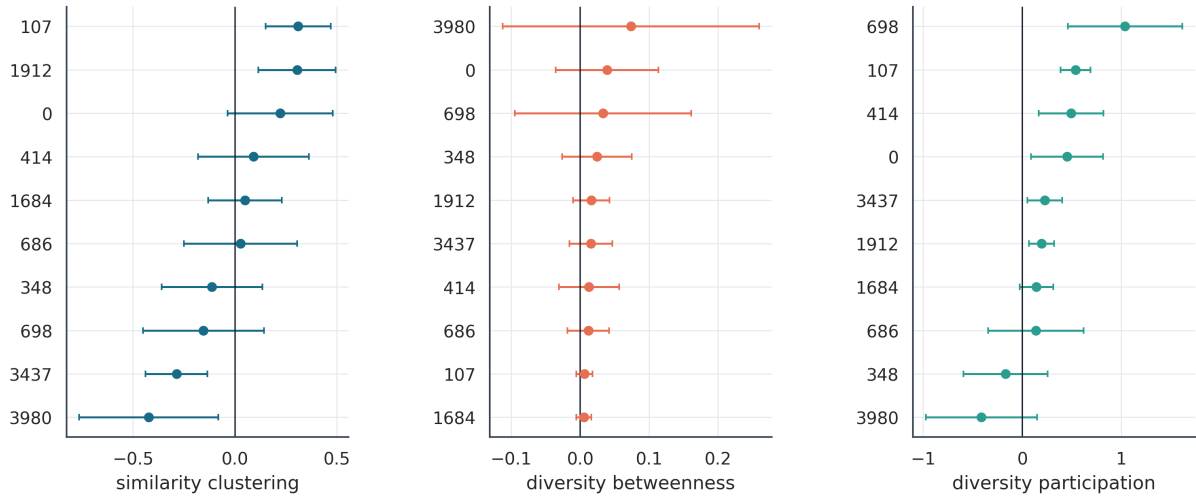


Figure 9: Heterogeneity across Facebook ego-networks

Notes: Each point reports the estimated coefficient within one ego-network. Horizontal lines are 95 % confidence intervals. The vertical line marks zero.

Discussion

The results provide stronger support for the bridging hypothesis than for the bonding hypothesis. Social diversity is positively associated with betweenness and, more clearly, with participation across communities. Both relationships remain after controlling for degree and ego-network fixed effects, and the diversity–betweenness and diversity–participation results are also supported by the permutation tests.

The evidence for bonding is weaker. Social similarity is positively related to local clustering in the adjusted model, but the effect is small and is not significant under the permutation test. Similarity may therefore contribute to local cohesion, but it does not fully explain it.

Sociologically, the findings suggest that having socially diverse contacts is linked to connecting different groups rather than simply having many friends. However, the effects vary across ego-networks, indicating that the broader structure of each network also matters.

One notable result is that the unconditional relationships were very weak. The expected patterns became clearer only after controlling for degree and differences across ego-networks. It was also surprising that the evidence for bridging was considerably stronger and more consistent than the evidence for bonding.

These conclusions are limited by the small number of ego-networks, the anonymized and incomplete profile attributes, and the absence of information on tie strength or interaction frequency. In addition, some anonymized users appear in more than one ego-network, which may introduce dependence across observations. Since the data are cross-sectional, the results should be interpreted as associations rather than causal effects.

Extensions and Future Research

A natural extension would be to use longitudinal network data to observe how changes in users’ social composition precede changes in cohesion or brokerage. This would help distinguish whether diverse contacts lead users to become brokers or whether users occupying brokerage positions subsequently form more diverse relationships.

A more causal design could exploit external shocks that alter opportunities for social interaction, such as entry into a university, workplace relocation, the creation of a new online group, or

platform feature rollouts. Users with different levels of exposure to these shocks could then be compared using a difference-in-differences or event-study design, provided that their network outcomes followed comparable pre-event trends and that no simultaneous event affected the groups differently.

Another possibility would be a randomized experiment in which users receive different friend or group recommendations. The effect of recommendations that increase social diversity could then be evaluated through subsequent changes in participation across communities, betweenness, and local clustering. These extensions would move the analysis from documenting structural associations toward identifying whether social diversity or similarity actually changes network position.

Conclusion

This study examined whether social similarity is associated with cohesion within groups and whether social diversity is associated with brokerage between groups in Facebook ego-networks. The results provide stronger evidence for the second relationship. Users with more socially diverse neighborhoods tend to connect a wider range of communities and occupy stronger brokerage positions, even after controlling for degree and differences across ego-networks. This pattern is especially clear for the participation coefficient and is also supported by the permutation tests.

The evidence for the bonding hypothesis is more limited. Social similarity is positively associated with local clustering in the adjusted model, but the effect is relatively small, sensitive to alternative specifications, and not significant in the corresponding permutation test. Similarity may contribute to cohesion, but it does not appear to be its main structural explanation in these networks.

Overall, the findings suggest that social diversity is more consistently linked to bridging between communities than social similarity is to cohesion within them. However, because the data are cross-sectional and limited to ten ego-networks, these results should be interpreted as structural associations rather than causal or population-wide conclusions about Facebook.

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